

SPLUNK TIER 1 SOC ANALYST

STUDENT WORKBOOK

Fill-in-the-blank exercises and lab worksheets for Lessons 1–15 and Labs 1–10.

This workbook contains no answers. Complete it during or after each lesson using your own Splunk instance and the BOTS v2 Attack-Only dataset (index=botsv2). Your instructor holds the separate answer key.

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Lesson 1 — The SOC Analyst Mindset & What Splunk Is

VOCABULARY CHECK — fill in the blank

1. A SOC is a team that watches an organization's systems for signs of attack.
2. A Tier 1 is the first responder who triages alerts and decides what's worth escalating.
3. A(n) Alert is a system-generated flag that a pattern of events looks worth a human's attention.
4. Triage is the process of quickly sorting alerts by urgency.
5. Escalation is the process of handing a confirmed or high-severity issue to a more senior analyst.

STUDENT EXERCISE

In your own words, write one sentence each defining: log, event, alert, triage, escalation.

Log: is a record of activity generated by a system or application

Event: is a single recorded occurrence of an activity

Alert: is a notification indicating that activity may require investigation

Triage: quickly evaluating and prioritizing an alert based on its severity and relevance

Escalation: passing an issue to a more senior analyst when it requires additional investigation or response

Lesson 2 — Installing & Verifying BOTS v2

VOCABULARY CHECK — fill in the blank

1. \$SPLUNK_HOME on Linux is /opt/splunk.
2. \$SPLUNK_HOME on Windows is C:\Program Files\Splunk.
3. BOTS v2 Attack-Only must be extracted into the Splunk etc/app directory.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 earliest =0  
| stats count
```

STUDENT EXERCISE

Run your completed search above. Record the total event count you see, and list the top 3 sourcetypes by count.

Total Event Count: 68,870,348

Top 3 Sourcetype by Count:

1. mysql:server:stats - 13,579,885
2. mysql:transaction:details - 13,574,451
3. Perfmon:Process - 11,868,094

Lesson 3 — Reading a Single Log Like an Analyst

VOCABULARY CHECK — fill in the blank

1. _time is the parsed timestamp field Splunk assigns to every event.
2. _raw contains the complete, unmodified original text of an event.
3. A(n) Field is a searchable name/value pair extracted from event data.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 earliest=0  
| head 5
```

STUDENT EXERCISE

Expand one event fully. Answer: What happened? When? Which host? Any user, IP, or process visible? Anything unusual?

Host: jabbah
User-agent: Mozilla/5.0 (Windows NT 6.333; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/59.0.3071.115 Safari/537.36
IP: 172.31.7.2
Site: store.froth.ly
/magento2/rest/default/V1/carts/mine/coupons/20twitter HTTP/1.1

Lesson 4 — SPL Fundamentals I — Filtering & Building Blocks

VOCABULARY CHECK — fill in the blank

1. In SPL, a bare space between two search terms means AND (both must be true).
2. The keyword OR means at least one of the grouped conditions must be true.
3. The keyword NOT excludes matching events.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="stream:http"  
(status=404 OR status=500)  
NOT status=200-
```

STUDENT EXERCISE

Using a sourcetype you discovered in Lab 1, build a search that filters to it AND excludes one common/benign value you notice.

The admin account may have generated failed authentication activity. I will filter Windows security events for admin, include failure-related events, and exclude successful events to narrow the investigation. So I use index=botsv2 sourcetype="wineventlog:security" admin* NOT success

Lesson 5 — Searching Like a SOC Analyst

VOCABULARY CHECK — fill in the blank

1. The analyst search loop, in order: start board → identify useful fields → narrow → summarize → investigate.
2. A(n) hypothesis is a plain-language guess to test, not an assumed fact.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 user=admin  
| stats count by host  
| sort - count
```

STUDENT EXERCISE

Write one search hypothesis in plain English about the BOTS v2 environment, then build SPL that actually tests it.

I suspect the admin user has unusually high activity on a particular host. I will test this by counting events for the admin user by host. When I use the SPL
index=botsv2 user=admin
| stats count by host
| sort - count

Lesson 6 — Time Is Everything

VOCABULARY CHECK — fill in the blank

1. earliest=0 = search from the beginning of indexed time (training-data-only usage).
2. The span parameter controls the bucket size in a timechart.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 earliest=0  
| timechart span=1h count
```

STUDENT EXERCISE

Run a timechart for one interesting sourcetype. Identify the busiest hour and explain what you'd check next before calling it suspicious.

I would run a timechart on one interesting sourcetype, such as wineventlog:security, to identify the busiest hour. I would then check the events during that hour to determine whether the activity is normal or associated with a specific user, host, or event type before calling it suspicious.
index=botsv2 sourcetype="wineventlog:security" earliest=0 | timechart span=1h count

Lesson 7 — Security Fields & Field Discovery

VOCABULARY CHECK — fill in the blank

1. The command fieldsummary lists every field present in a result set along with counts and top values.
2. A good analyst does not memorize every field — they know how to discover them.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="<sourcetype>"  
| fieldsummary
```

STUDENT EXERCISE

Run fieldsummary on two different sourcetypes. List three fields present in one but absent in the other.

I compared wineventlog:security and stream:http using fieldsummary. Three fields that may be present in one but absent in the other are user, src_ip, and status, because different sourcetypes contain different types of telemetry.

SPL

```
index=botsv2 sourcetype="wineventlog:security" | fieldsummary
```

```
index=botsv2 sourcetype="stream:http" | fieldsummary
```

Lesson 8 — Windows Telemetry Investigation

VOCABULARY CHECK — fill in the blank

1. Windows telemetry can reveal: authentication, account activity, process creation, and privilege activity.
2. High account activity is is not (circle one) automatically suspicious.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="<win sourcetype>"
| stats count by user
| sort - count
```

STUDENT EXERCISE

Identify one high-activity account. Build a table of its events with _time, host, user, and action (or the equivalent field you discovered).

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Lesson 9 — Network Log Analysis

VOCABULARY CHECK — fill in the blank

1. The five core network questions: Who initiate? Where did it go? What port? How often? What protocol?

BUILD THE SPL — fill in the missing pieces

```
index=botsv2
| stats count by src_ip dest_ip dest_port
| sort - count
```

STUDENT EXERCISE

Identify one dest_ip that stands out. Investigate which host(s) and user(s) were involved.

The destination IP that stands out is 172.31.38.181 with 588,786 events. I would investigate which source IPs and users communicated with this destination, along with the destination port and protocol, to determine what traffic is responsible for the high volume and then I use this SQL

```
index=botsv2 dest_ip="172.31.38.181"
| stats count by dest_port, host
| sort - count
```

to investigate which hosts and users were involved while I can't use if the users were involved since it shows no results yet but for host there only one host: matar

Lesson 10 — DNS Analysis

VOCABULARY CHECK — fill in the blank

1. The golden DNS rule: unfamiliar domain _____ malicious.
2. Regular, fixed-interval queries from one host may suggest _____ beaoning _____.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="<dns sourcetype>" "  
| stats count by query  
| sort - count | head 20
```

STUDENT EXERCISE

Pick one non-obvious domain from your top-20 list. Investigate the host, frequency, and time window.

- Domain: store.froth.ly
- Host: cassiopeia
- Frequency: 1,518 queries
- First seen: 08/04/2017 02:04:16
- Last seen: 08/29/2017 19:09:58
- Pattern: Repeated DNS queries over an extended period.
- Assessment: Suspicious / requires further investigation, because the domain is non-obvious and queried frequently. The data alone does not prove it is malicious.

The domain store.froth.ly was queried 1,518 times by host cassiopeia from August 4 to August 29, 2017. The repeated activity over an extended period warrants investigation for possible automated or beaoning behavior.

Lesson 11 — HTTP / Web Analysis

VOCABULARY CHECK — fill in the blank

1. A spike in _____ 404 _____ responses from one source often suggests enumeration/scanning.
2. Scanning behavior looks like: many distinct paths, high _____ error _____ rate, low repeat.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 sourcetype="<web sourcetype>" status=400  
| stats count by uri  
| sort - count
```

STUDENT EXERCISE

Find one src_ip requesting an unusually high number of distinct URIs. Inspect what those paths are.

```
SQL: index=botsv2  
| stats count by http_method uri_path status  
| sort - count
```

Then investigate the suspicious/unusual URI with: index=botsv2 uri_path="<URI_PATH>"
| table _time host src_ip dest_ip http_method uri_path status
| sort _time

- URI/path: <URI_PATH>
- HTTP method: <METHOD>
- Status: <STATUS>
- Host/source: <HOST or SRC_IP>
- Why it stands out: Unusual or repeated HTTP request activity that warrants further investigation.

Lesson 12 — Stats Mastery — sort, dedup, top, eval

VOCABULARY CHECK — fill in the blank

1. count = counts total events; dc (field) counts unique values of that field.
2. The dedup command removes duplicate events, keeping one per unique field value.
3. The eval command creates or transforms a field.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2
| stats earliest(_time) as first latest(_time) as last by host
| eval span=last-first
```

STUDENT EXERCISE

Using eval, calculate the activity time span (first-seen to last-seen) for one host of your choice.

Host: jabbah

First seen: 1501743777.632071

Last seen: 1504149431.304235

Activity span: 2405653.672164 seconds (approximately 27 days, 20 hours, 14 minutes, 14 seconds)

SPL Used:

```
index=botsv2 | stats earliest(_time) as first latest(_time) as last by host | eval span=last-first | sort -span
```

Lesson 13 — Correlation — Thinking Across Data Sources

VOCABULARY CHECK — fill in the blank

1. Correlation chain, in order: IP → Host → User → Process → Network connection → Domain → Timestamp.

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 dest_ip="<suspicious id>"
| stats values(src_ip) values(src_ip) by dest_ip
```

STUDENT EXERCISE

Starting from the suspicious IP identified in Lab 2, build a cross-sourcetype timeline for the host involved, covering ±30 minutes around the activity.

172.31.38.181 → 111.249.38.34 → source port 25 → 2 events

The suspicious IP 111.249.38.34 was associated with host `matar`. On August 19, 2017 at approximately 22:04:33, host `matar` communicated from 172.31.38.181 to 111.249.38.34. The activity included source port 25 and destination port 3162, with one event identifying TCP. No user, process, or domain information was populated in the available events. The suspicious IP was observed in 2 events.

Lesson 14 — Detections, Alerts & Tier 1 Triage

VOCABULARY CHECK — fill in the blank

1. A(n) detection is defined suspicious-behavior logic; a(n) alert is that logic scheduled to notify someone.
2. The 10-step triage workflow begins: Alert → Validate → Identify affected asset → Identify user → ...

BUILD THE SPL — fill in the missing pieces

```
index=botsv2
| stats count by src_ip
| where count > 20
```

STUDENT EXERCISE

Pick one `src_ip` from the detection output above. Run a full 10-question triage and decide escalate/close.

Using your selected IP 111.249.38.34

Selected IP: 111.249.38.34

Events: 104,984

Host: `matar`

User: Not identified

Source: 172.31.38.181

Destination: 111.249.38.34

Ports: 25 → 3162

Protocol: TCP

Time: 2017-08-19 22:04:33

Activity: Repeated

Severity/Scope: Requires further investigation

Decision: Escalate

Lesson 15 — Timelines, Reporting & the Final Capstone

VOCABULARY CHECK — fill in the blank

1. The Tier 1 report field list includes: Alert, Time, Affected Host, User, Source IP, Destination IP, Observed Activity, Evidence, Assessment, Severity, Recommended Action, and Escalation.
2. Rule: write "the evidence indicates..." rather than "this is definitely malware."

BUILD THE SPL — fill in the missing pieces

```
index=botsv2 host="<host>" earliest="<start>" latest="<end>"
| table _time sourcetype user src_ip dest_ip action process command
| sort _time
```

STUDENT EXERCISE

Complete the reporting template for any prior lab using only evidence-based language.

Alert: Activity involving 111.249.38.34 exceeded the detection threshold.
Time: 2017-08-19 22:04:33.017–22:04:33.185
Affected Host: matar
User: Not identified
Source IP: 172.31.38.181
Destination IP: 111.249.38.34
Observed Activity: Network communication recorded as `stream:ip` and `stream:tcp`.
Evidence: Two events show communication from 172.31.38.181 to 111.249.38.34.
Assessment: The evidence indicates network communication between the source and destination IPs on host matar.
Severity: Requires further investigation.
Recommended Action: Investigate the network activity and gather additional user, process, and domain context.
Escalation: Escalate.

Lab Worksheets 1–10

Complete each lab using your own BOTS v2 environment. Fill in every blank in the guided steps before running the search, then answer the investigation questions in the space provided.

Lab 1 — Find Your Data

Scenario: You are a newly hired Tier 1 Analyst. Your first task is to understand what telemetry is available in the SIEM before investigating anything.

GUIDED STEPS — fill in the blanks

```
index=botsv2 earliest=0
| stats count

index=botsv2 earliest=0
| stats count by sourcetype
| sort - count

index=botsv2 earliest=0
| stats count by host
```

INVESTIGATION QUESTIONS

1. How much data exists in total? 68,870,348
2. How many distinct sourcetypes are present? 103
3. Which sourcetype contains the most events? mysql:server:stats (13579885)
4. What categories of security telemetry appear to be available (Windows, network, web, DNS, etc.)?

Write - Windows/endpoint telemetry, authentication and account activity, process activity, network logs, DNS, and HTTP/web telemetry appear to be available.

Escalation decision - No immediate escalation based solely on the volume of data. Continue with telemetry discovery and investigate individual suspicious events or patterns before escalating.

Analyst conclusion (benign / suspicious / malicious, with evidence):

The environment contains a large and diverse security dataset with approximately 68.9 million events and 103 sourcetypes. The available telemetry includes endpoint, authentication, network, DNS, and web-related data, providing multiple sources for security investigation.

Escalation decision:

No immediate escalation based solely on the volume of data. Continue with telemetry discovery and investigate individual suspicious events or patterns before escalating.

Lab 2 — Investigate a Suspicious IP

Scenario: Threat intelligence reports a suspicious IP address. Your task is to determine whether systems in the environment communicated with it.

GUIDED STEPS — fill in the blanks

```
index=botsv2
| stats count by src_ip
| sort - count (use this to pick a candidate IP)

index=botsv2 src_ip=" <candidate> "

index=botsv2 src_ip=" <candidate> "
| stats count by src_ip dest_ip
```

INVESTIGATION QUESTIONS

1. Is this an isolated event or repeated behavior? it is repeated behavior because the IP is associated with many events in the BOTSv2 environment, indicating repeated activity rather than an isolated event.
2. Which host and user are tied to this activity? The activity is associated with the host `matar`. No user value was identified in the search results.
3. What time window does the activity span? The activity occurred from August 19, 2017 at 12:52:23 UTC to August 19, 2017 at 14:04:33 UTC, spanning approximately 1 hour and 12 minutes.
4. What port and protocol are involved? The activity involves source IP 172.31.38.181 communicating with destination IP 111.249.38.34 using source port 25. The protocol is not displayed in the current result.

Analyst conclusion (benign / suspicious / malicious, with evidence):

The activity is repeated and is associated with host `matar`. The connection to `111.249.38.34` involves port 25 and TCP activity. However, the available evidence does not identify a user, process, or domain, so additional context should be reviewed before determining whether the activity is benign, suspicious, or malicious.

Escalation decision:

Continue investigation and consider escalation because the activity is repeated and involves an external destination and network port 25. Review additional indicators such as the process, user, domain, and surrounding events before making a final determination.

Lab 3 — Suspicious User Activity

Scenario: The SOC received a report that a user account may have been compromised.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_<authentication_sourcetype>_" user="*username"
| table _time host user src_ip action
| sort _time
```

INVESTIGATION QUESTIONS

1. Where did the user authenticate from?
2. When did the activity occur?
3. Which systems were involved?
4. Was there suspicious activity immediately after login?
5. Does the overall pattern look normal for this account?

Analyst conclusion (benign / suspicious / malicious, with evidence):

Escalation decision:

Lab 4 — DNS Investigation

Scenario: A workstation is generating unusual DNS activity.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_<sourcetype>_"
| stats count by query{}
| sort - count
```

INVESTIGATION QUESTIONS

1. Which domain(s) stand out, and why? The domain/query that stands out is FHFAEBEECACACACACACACACACACAAA because it generated 13,937 DNS queries, making it the most

frequently queried unusual-looking name in the results. Its encoded-looking format also differs from common legitimate domains such as Microsoft and Google services.

2. Which host queried it, and how often? The host `jupiter` queried `FHFAEBEECACACACACACACACACACAAA` 7,371 times.
3. Is the query timing regular/periodic, or irregular? Irregular. The DNS query timing is irregular because the intervals between queries vary widely, ranging from milliseconds to several minutes or much longer gaps.
4. What corroborating evidence (network/HTTP) supports or weakens your hypothesis? The DNS activity supports the hypothesis because the host `jupiter` repeatedly queried the unusual domain `FHFAEBEECACACACACACACACACACAAA` 7,371 times. The irregular timing of the queries also makes the activity noteworthy. However, no HTTP/network connection evidence has been provided yet, so there is not enough corroborating evidence to confirm the hypothesis.

Analyst conclusion (benign / suspicious / malicious, with evidence):

It is suspicious because the host `jupiter` generated 7,371 DNS queries for the unusual query `FHFAEBEECACACACACACACACACACAAA`. The query timing is irregular, with intervals ranging from fractions of a second to very long gaps. This pattern warrants further investigation, although the available evidence does not by itself confirm malicious activity.

Escalation decision:

Investigate the DNS activity further and correlate it with network/HTTP traffic and other host activity.

Lab 5 — Build a Basic Detection

Scenario: You have been asked to create a basic detection for unusually high connection activity.

GUIDED STEPS — fill in the blanks

```
index=botsv2
| stats count by src_ip
| sort - count

index=botsv2
| stats count by src_ip
| where count > 20
```

INVESTIGATION QUESTIONS

1. How many `src_ips` match at threshold `> 10`? 175
2. How many match at `> 20`? 162
3. How many match at `> 50`? 152
4. Which threshold would you recommend, and why? Threshold `> 20` because it provides a reasonable balance between detecting unusually high connection activity and limiting excessive alerts. It identifies 162 `src_ips`, while the higher `> 50` threshold identifies only 152.

Analyst conclusion (benign / suspicious / malicious, with evidence):

The results show significant high connection activity, with 173 `src_ips` exceeding 10 connections, 162 exceeding 20, and 152 exceeding 50. This indicates that several source IPs have unusually high

activity and should be investigated further. High connection volume alone does not confirm malicious activity.

Escalation decision:

Investigate the source IPs with high connection counts and correlate them with destination IPs, ports, protocols, hosts, users, and other security events.

Lab 6 — Brute-Force Authentication Investigation

Scenario: Multiple failed logins have been reported against one account.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype=" <win sourcetype> "  
| stats count by user, _src_ip, action  
| where action="failure" AND count > 10
```

INVESTIGATION QUESTIONS

1. Do the failures come from a single source or many?
2. Is a failure streak followed by a success?
3. What MITRE ATT&CK technique would you consider mapping this to, and why only after reviewing evidence?

Analyst conclusion (benign / suspicious / malicious, with evidence):

Escalation decision:

Lab 7 — Suspicious PowerShell Activity

Scenario: An analyst suspects PowerShell misuse on a host.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="Powershell:ScriptExecutionSummary"  
| search process name="*powershell"
```

INVESTIGATION QUESTIONS

1. Is there evidence of an encoded command or download cradle? No evidence of an encoded command or download cradle was found in the examined events. The PowerShell command lines only show splunk-powershell.exe and --ps2.

2. What is the parent process of the PowerShell execution? The parent process is C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe, running as a service. But it won't show up in the statistics
3. Is this use consistent with normal admin activity, or does it look abnormal? Why? The activity appears consistent with normal Splunk administrative/service activity because splunk-powershell.exe is launched by the Splunk splunkd.exe service under NT AUTHORITY\SYSTEM, with no suspicious encoded or download-related arguments observed.

Analyst conclusion (benign / suspicious / malicious, with evidence):

The PowerShell activity is associated with the Splunk Universal Forwarder. The parent process is splunkd.exe, the execution runs as NT AUTHORITY\SYSTEM, and the observed command lines do not contain encoded-command or download-cradle indicators.

Escalation decision:

Do not escalate based on these events alone. The observed PowerShell execution is consistent with Splunk Universal Forwarder service activity and no clear malicious indicators were identified.

Lab 8 — Network Reconnaissance

Scenario: Possible port or host scanning is suspected.

GUIDED STEPS — fill in the blanks

```
index=botsv2
| stats dc(dest_port) by _src_ip_
| sort - dc(dest_port)

index=botsv2
| stats dc(_dest_ip_) by src_ip
| sort - dc(_dest_ip_)
```

INVESTIGATION QUESTIONS

1. Which src_ip touches the widest breadth of ports or destinations? 10.0.1.120 touches the widest breadth of ports with 5,769 distinct destination ports. 10.0.2.107 touches the widest breadth of destinations with 566 distinct destination IPs.
2. Does this pattern look like scanning rather than normal usage? Why? Yes, the unusually high number of distinct ports and destinations is consistent with possible network reconnaissance or scanning. However, additional investigation is needed to determine whether the activity is malicious or legitimate.

Analyst conclusion (benign / suspicious / malicious, with evidence):

10.0.1.120 contacted 5,769 distinct destination ports, while 10.0.2.107 contacted 566 distinct destination IPs. The unusually broad network activity is consistent with possible scanning, but the statistics alone do not establish malicious intent.

Escalation decision:

Escalate for further investigation. Examine the highest-volume source IPs, destination IPs/ports, timestamps, protocols, and connection patterns to determine whether the activity represents legitimate network behavior or reconnaissance.

Lab 9 — Web Attack Investigation

Scenario: Possible web exploitation attempt against an internet-facing system.

GUIDED STEPS — fill in the blanks

```
index=botsv2 sourcetype="_access_combined_"
| stats count by _src_ip_ uri
| sort - count
```

INVESTIGATION QUESTIONS

1. Are there unusual HTTP methods or status-code patterns? Yes. Most requests use GET/POST, but unusual methods such as PROPFIND, SEARCH, TRACE, OPTIONS, PUT, and USER were observed. There were also many HTTP 404 responses (4,049 GET requests), indicating numerous requests for nonexistent resources.
2. Do any requested URIs look like exploitation attempts (e.g., admin panels, config files)? Yes. There were numerous attempts to access administrative and phpMyAdmin paths, including /phpmyadmin/, /admin/phpmyadmin/, and /db/phpmyadmin/. More significantly, a request to gacrux targeting /admin/index.php?module=user-titles&action=edit... contained URL-encoded JavaScript with a <script> payload and an XMLHttpRequest intended to submit data to a user-creation endpoint. This is consistent with an attempted web application exploit/XSS-style injection.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Suspicious. The logs show automated probing for administrative/phpMyAdmin interfaces and a crafted JavaScript injection payload targeting the administrative application. The payload attempts to send a request to create a user account, providing evidence of an exploitation attempt.

Escalation decision:

Escalate for investigation. Review the affected gacrux host, determine whether the injected request was successful, identify the source IP associated with the activity, and check for any newly created or modified accounts and subsequent activity.

Lab 10 — Multi-Stage Timeline & Full Triage

Scenario: Combine findings from Labs 2, 4, and 8 into a single cross-sourcetype timeline and run a full Tier 1 triage.

GUIDED STEPS — fill in the blanks

```
index=botsv2 host="_matar_" earliest="_08/19/2017:12:52:23_"
latest="_08/19/2017:14:04:33_"
| table _time sourcetype user src_ip dest_ip action
| sort _time
```

INVESTIGATION QUESTIONS

1. What story does the chronological sequence tell? jupiter shows repeated unusual DNS activity and broad network communication involving 10.0.2.107 and 10.0.1.120. matar separately shows repeated external TCP/25 communication with 111.249.38.34. These are suspicious observations, but the current evidence does not prove they are one coordinated attack chain.
2. What is your final severity assessment? Suspicious because it requires further investigation. There are several indicators worth investigating: unusual repeated DNS activity, broad internal network communication, and repeated external TCP/25 communication.
3. Escalate or close — and why? Escalate for further investigation. The activity should not be closed as benign based on the current evidence. Additional correlation with process creation, users, DNS details, HTTP activity, and authentication/account activity is needed to determine whether the behavior was legitimate or malicious.

Analyst conclusion (benign / suspicious / malicious, with evidence):

Suspicious — jupiter showed repeated unusual DNS queries (FHFAEBEECACACACACACACACACACAAA) and broad network activity involving 10.0.2.107 and 10.0.1.120. matar also showed repeated external TCP/25 communication with 111.249.38.34. These behaviors warrant investigation, but the available evidence does not confirm malicious activity or a single coordinated attack.

Escalation decision:

Escalate for further investigation — correlate the activity with process execution, users, DNS/HTTP details, authentication events, and additional network telemetry to determine whether the behavior is legitimate or malicious.